

Introduction to Radiographs & Dental X-Ray Film

Radiograph: An image of an object produced using X-rays instead of light.

Dental X-Ray Film: The recording medium on which the image is formed after being exposed to X-rays.

Types of X-Ray Film:

- a. Intraoral X-Ray Film: Placed inside the mouth.
- b. Extraoral X-Ray Film: Placed outside the mouth.

1. Intraoral X-Ray Film

Chemical Composition & Structure

Intraoral film is a complex packet designed to capture a clear image while protecting the film.

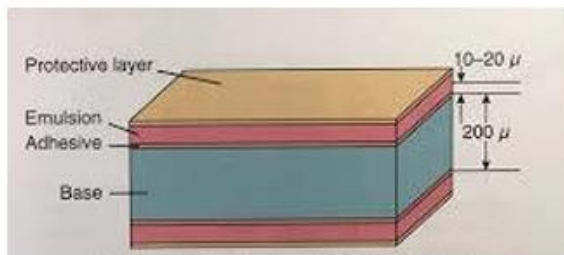
Film Packet Contents:

- **Outer Wrapper:** A moisture-resistant, light-proof plastic or paper envelope.
- **Lead Foil:** Located behind the film. It absorbs scattered (secondary) radiation, reducing film fog and minimizing patient exposure to radiation that has already passed through the film. It often has a distinctive herringbone pattern.
- **Black Paper:** Wraps the film, protecting it from light and mechanical friction.
- **The Film Itself:** The core component where the image is recorded.

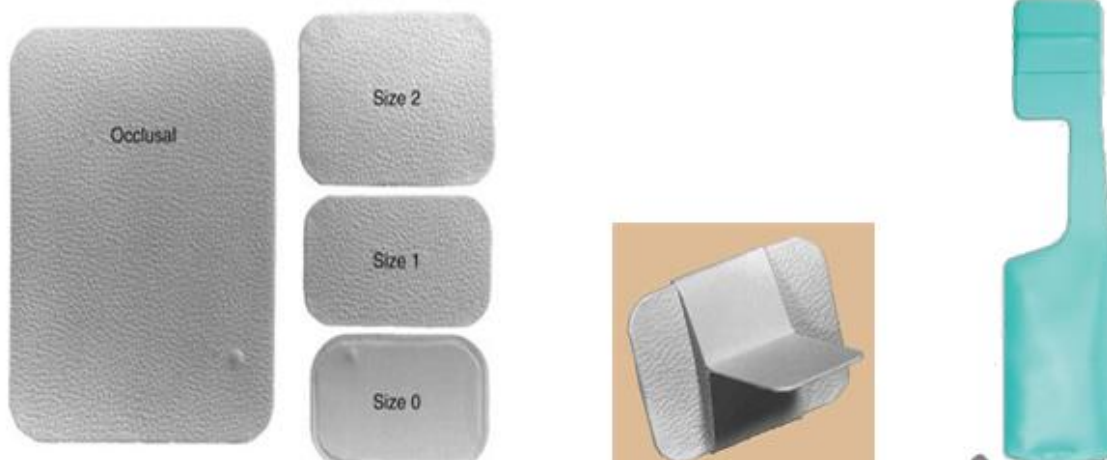
Film Structure:

- **Base:** A flexible, transparent, and fracture-resistant foundation made of polyester or cellulose acetate.
- **Emulsion:** A gelatin layer coated on both sides of the base, containing light- and radiation-sensitive crystals of Silver Halide (mainly Silver

Bromide, AgBr). These crystals are the "heart" of the film, capturing the latent image.



Types of Intraoral Film



Intraoral films are classified numerically based on their size and purpose.

1. Type I - Periapical Film:

Purpose: Examines the entire tooth, from crown to root tip, and the surrounding bone.

Sizes:

Size 0: For small children (22mm x 35mm).

Size 1: For anterior teeth in adults (24mm x 40mm).

Size 2: Standard size for adults (30.5mm x 40.5mm).

2. Type II - Bitewing Film:

Purpose: Detects decay (caries) between teeth and monitors the bone levels between adjacent teeth. Features a "wing" tab for the patient to bite on.

Sizes: Sizes 0, 1, 2 (as above), and a longer Size 3 for posterior teeth.

3. Type III - Occlusal Film:

Purpose: Captures a larger area of the jaw (e.g., an entire arch or a section of it) than a periapical film.

Size: Larger, typically only one size (e.g., 57mm x 76mm).

Film Speed

Definition: The sensitivity of the film's silver halide crystals to radiation. A faster film requires less radiation to produce an image, reducing patient dose.

Key Principle: Larger silver halide crystals result in faster film speed.

Classification: Rated alphabetically from A (slowest) to F (fastest). Only D-speed or faster films are used intraorally.

D-Speed (Ultra-Speed): Older standard.

E/F-Speed (Insight): Modern standard; requires about half the exposure time and radiation dose of D-speed film.

[Image Suggestion: A chart comparing the exposure time or required radiation dose for D-speed vs. E/F-speed film.]

2. Extraoral X-Ray Film

Used for larger views like panoramic or cephalometric X-rays. They are used in conjunction with cassettes.

1. Non-Screen Film:

The film emulsion is directly sensitive to X-rays.

Has a thicker emulsion for greater sensitivity.

Used without intensifying screens.

2. Screen Film:

The film emulsion is sensitive to visible light (typically blue or green).

Used inside a cassette with Intensifying Screens. These screens contain phosphors (e.g., Calcium Tungstate) that fluoresce (emit light) when struck by X-rays. This light, in turn, exposes the film.

Advantage: Significantly reduces the radiation dose required because the screens are more efficient at converting X-rays to light than the film is at converting X-rays directly to an image.



Film Properties

The quality of a radiographic image is defined by three key properties:

A. Density



Definition: The overall blackness or darkness of the processed film.

Measurement: Measured as Optical Density (OD) using a densitometer. The relationship between exposure and density is shown by the H & D (Hurter and Driffield) Curve.

Components:

Base Density: Inherent slight darkness of the film base (~ 0.1 OD).

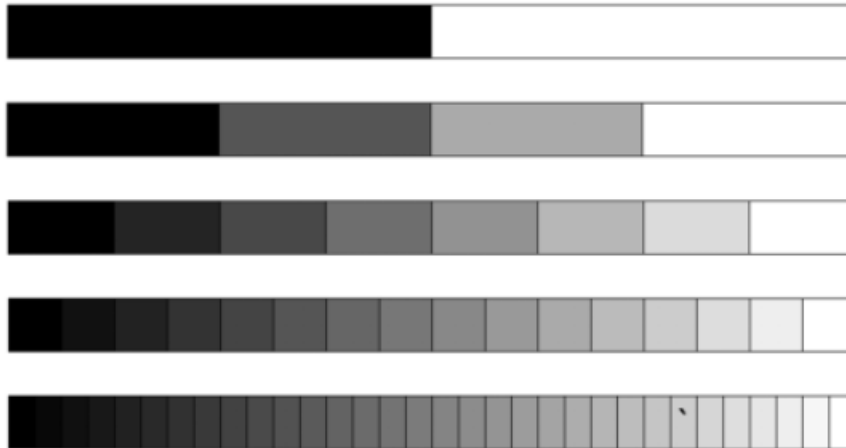
Fog Density: Unwanted density from storage, improper processing, or stray radiation (should be < 0.2 OD).

Factors Increasing Density:

Increase in Exposure time, Increase in milliamperage (mA), Increase in kilovoltage (kVp)

Increase in Developing time, Decrease in Distance from X-ray tube to film

B. Contrast





Definition: The difference in densities between various areas on a radiograph.

Types:

High Contrast (Short Scale): Sharp black-and-white differences with few shades of gray. Result of lower kVp.

Low Contrast (Long Scale): Many shades of gray, showing more subtle detail. Result of higher kVp.

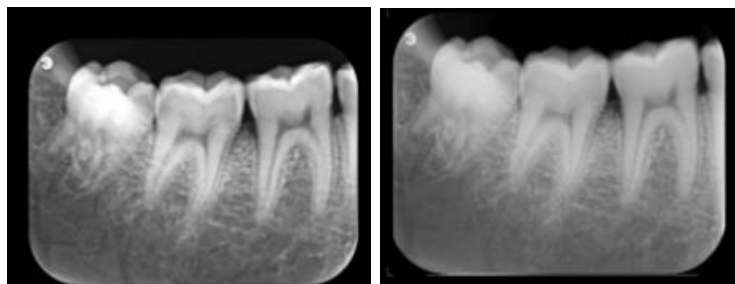
Factors Affecting Contrast:

Kilovoltage (kVp): The primary controlling factor. Increasing kVp increases penetration and creates a longer scale of contrast (more grays).

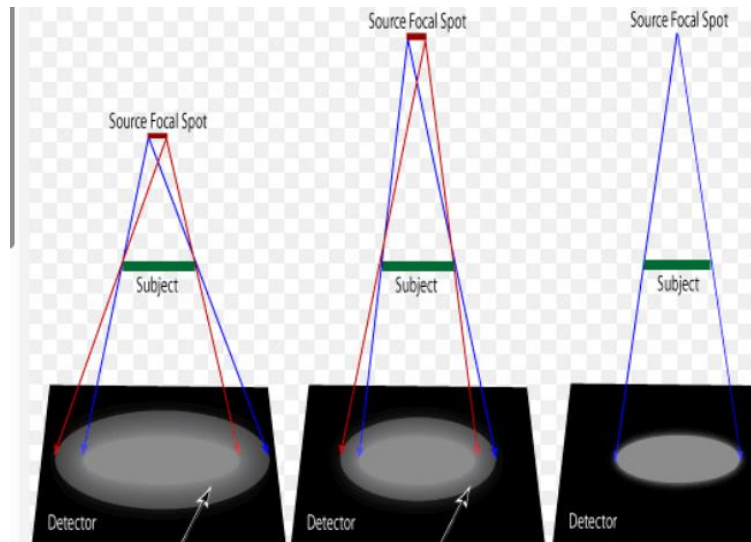
Processing: Increased developer temperature can decrease contrast.

C. Detail/Definition

Definition: The sharpness and clarity of outlines of structures in the image.



Factors Affecting Detail:



Focal Spot Size: Smaller is sharper.

Film Grain Size: Smaller silver halide crystals produce sharper images.

Movement: Any movement of patient, tube, or film causes blurring.

Geometry:

Target-Object Distance: Should be as long as possible.

Object-Film Distance: Should be as short as possible.

Screen-Film Contact: Poor contact in cassettes causes blurring.

Image Formation & Processing

Latent Image Formation

When X-ray photons strike the AgBr crystals, they create an invisible change—a "latent image." This involves the formation of microscopic specks of metallic silver at sensitivity sites within the crystal. This image remains invisible until chemical processing.

Film Processing Steps

Processing converts the latent image into a visible, permanent radiograph.

1. Developing:

Converts the latent image to a visible one.

The film is placed in an alkaline developer solution, which reduces the exposed AgBr crystals to black, metallic silver.

2. Rinsing:

A water rinse (approx. 30 seconds) stops the development action and removes excess developer.

3. Fixing:

The film is placed in an acidic fixer solution which:

Removes the unexposed/undeveloped AgBr crystals, clearing the film.

Re-hardens the softened emulsion.

4. Washing:

A thorough wash in running water removes all processing chemicals to ensure archival stability.

5. Drying: The film is air-dried.

The Darkroom

A light-tight room for manual film handling and processing.

Size: Minimum of 20 sq. ft. for an average dental office.

Location: Should be convenient, cool, with minimal temperature fluctuation, low humidity, and access to plumbing and power.

Illumination:

White Ceiling Light: For general illumination when no films are out.

Safelight: A filtered light (usually red or amber) that provides dim illumination safe for handling unprocessed film for short periods. It uses a low-wattage bulb and must be placed at the correct distance.

Warning Light: A red light outside the door indicating when the darkroom is in use.

Film Storage

Store films in a cool, dry place away from chemicals.

Protect from pressure to avoid artifacts.

Use lead-lined or steel storage boxes to protect films from stray radiation fog.